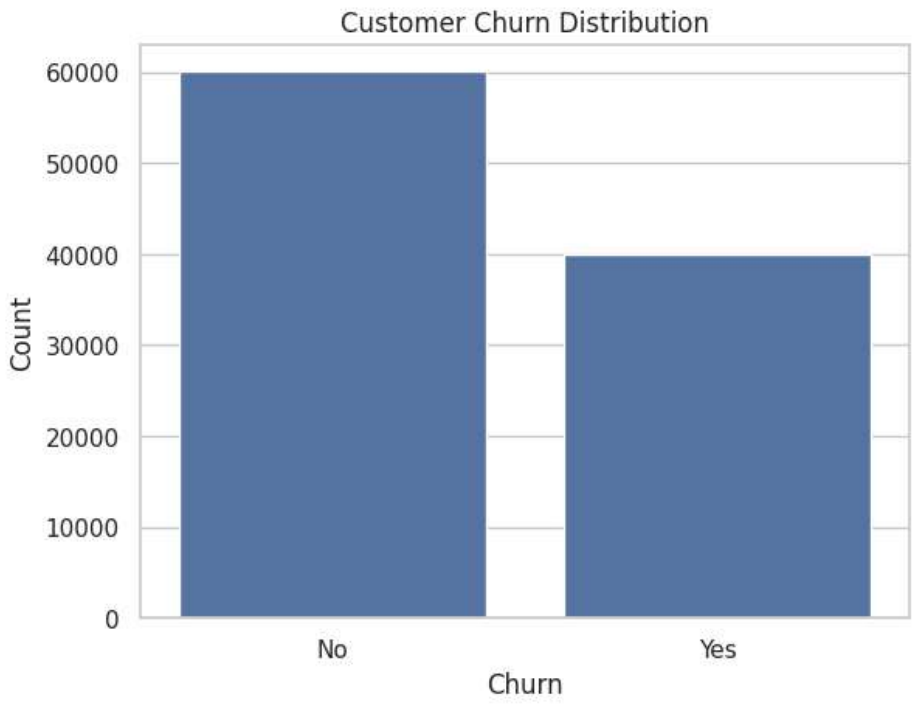
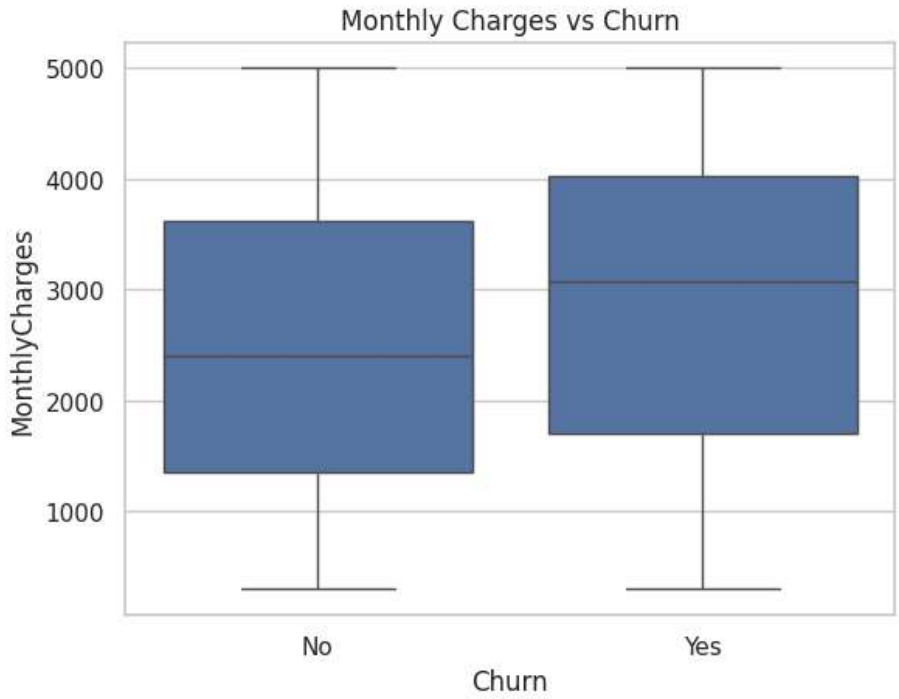
DATA VISUALIZATION

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")
```

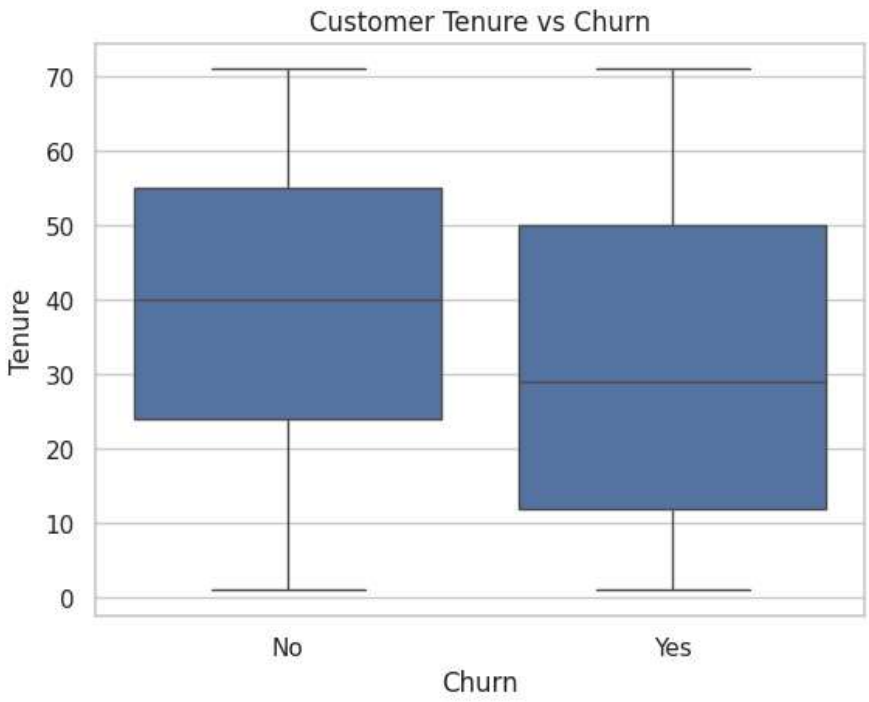
```
import pandas as pd
df = pd.read_csv('/content/large_customer_churn_dataset.csv')
sns.countplot(x="Churn", data=df)
plt.title("Customer Churn Distribution")
plt.xlabel("Churn")
plt.ylabel("Count")
plt.show()
```



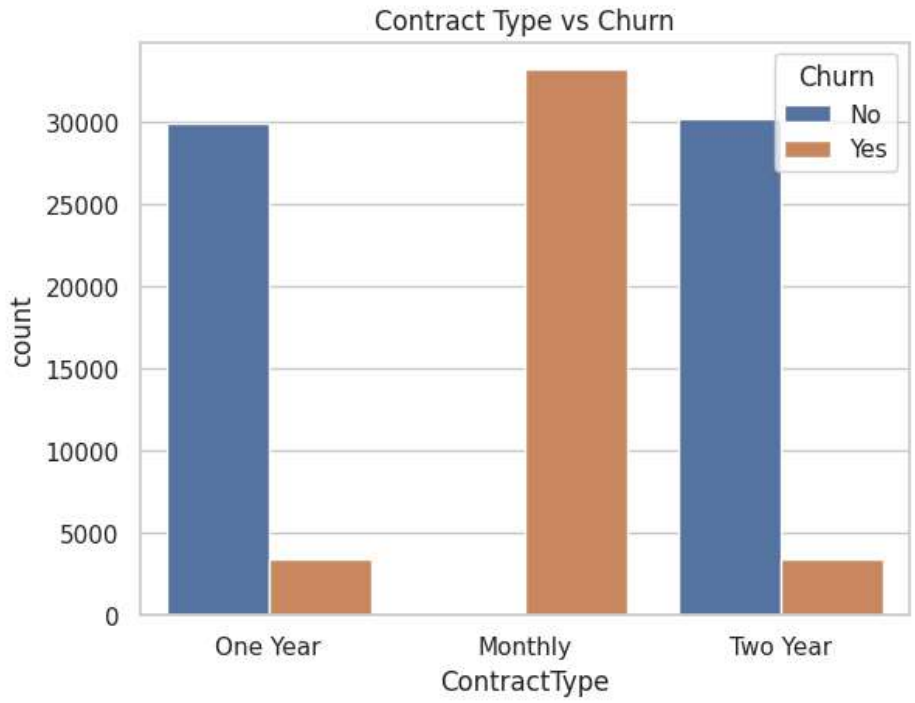

```
sns.boxplot(x="Churn", y="MonthlyCharges", data=df)
plt.title("Monthly Charges vs Churn")
plt.show()
```



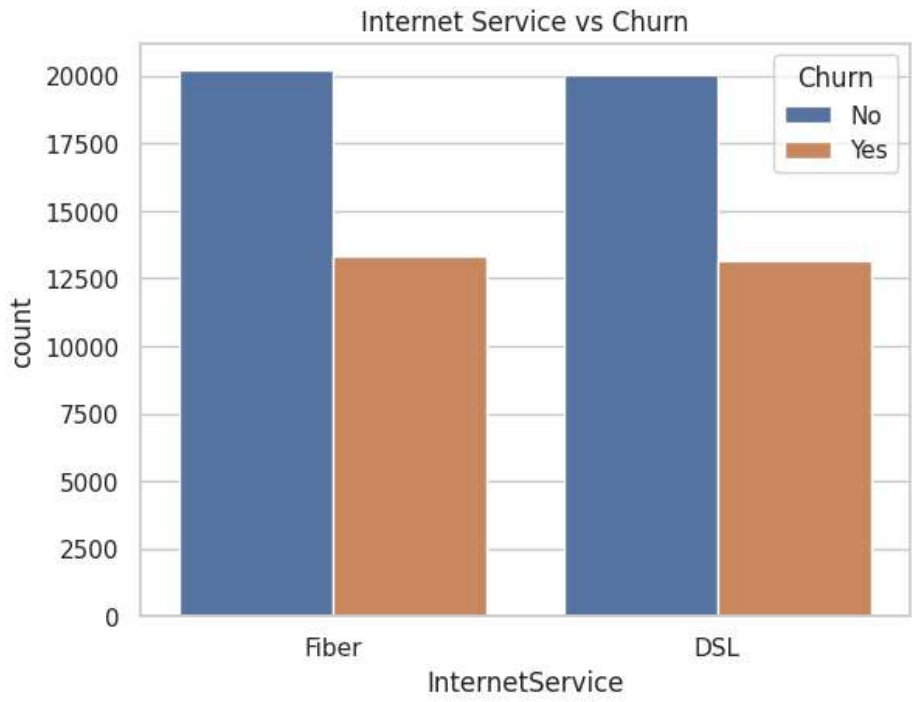

```
sns.boxplot(x="Churn", y="Tenure", data=df)
plt.title("Customer Tenure vs Churn")
plt.show()
```



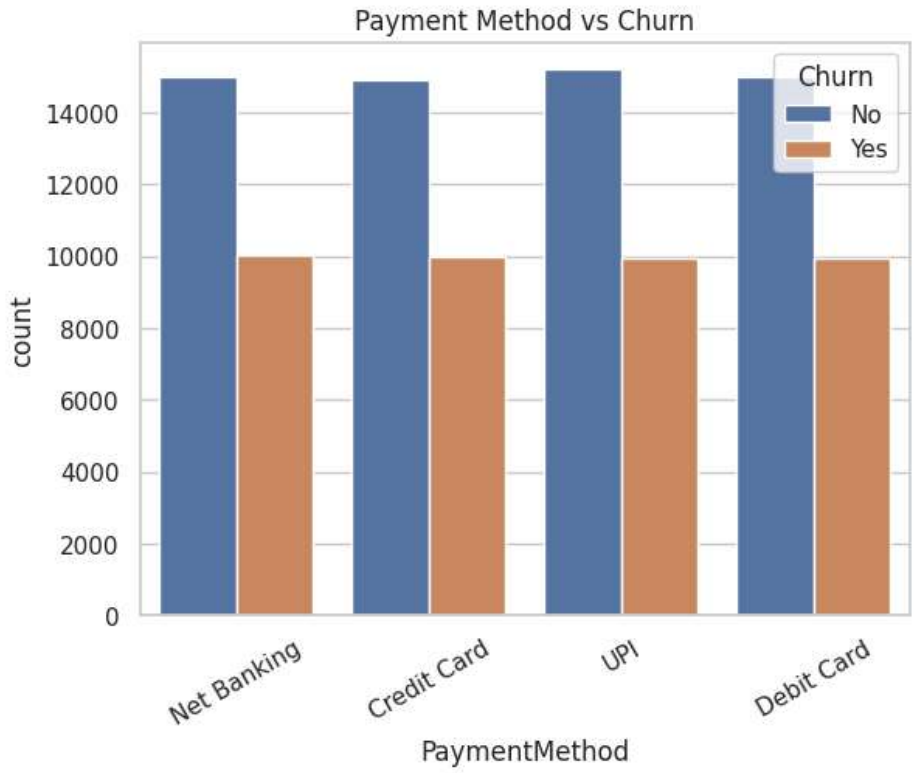

```
sns.countplot(x="ContractType", hue="Churn", data=df)
plt.title("Contract Type vs Churn")
plt.show()
```



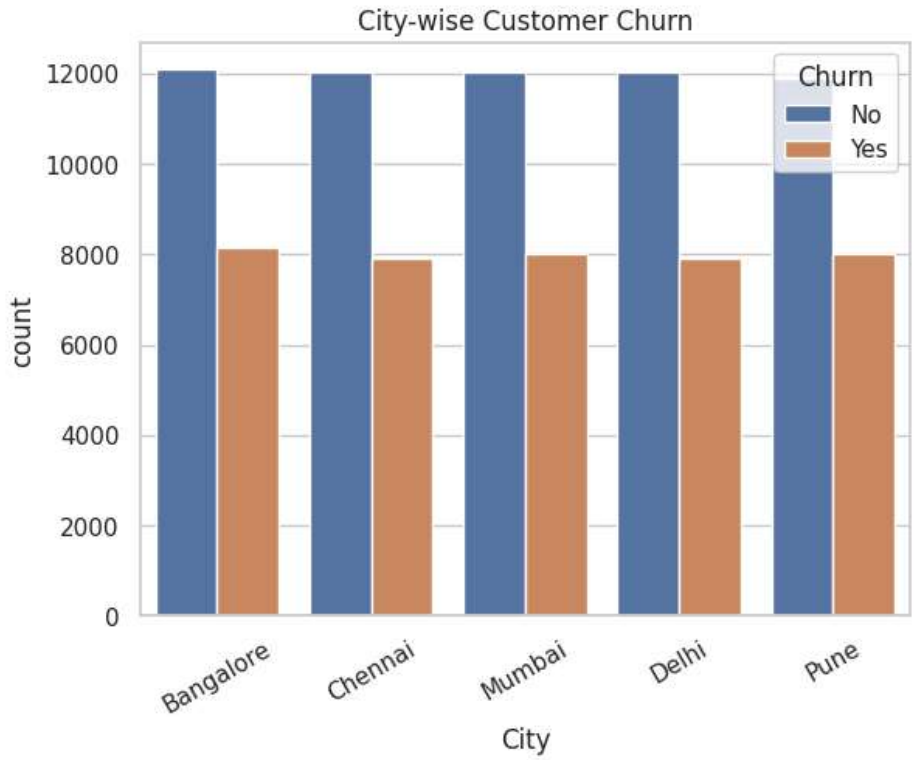

```
sns.countplot(x="InternetService", hue="Churn", data=df)
plt.title("Internet Service vs Churn")
plt.show()
```



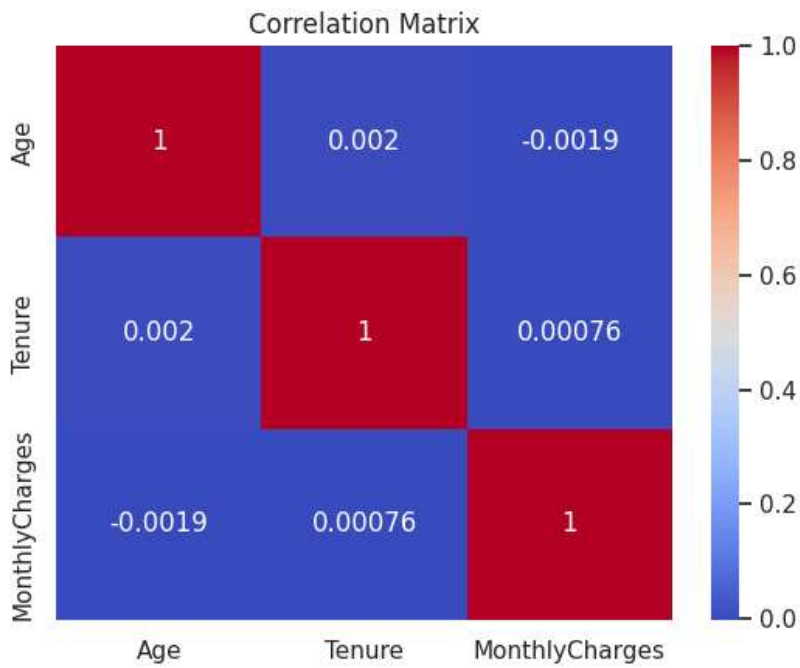

```
sns.countplot(x="PaymentMethod", hue="Churn", data=df)
plt.title("Payment Method vs Churn")
plt.xticks(rotation=30)
plt.show()
```



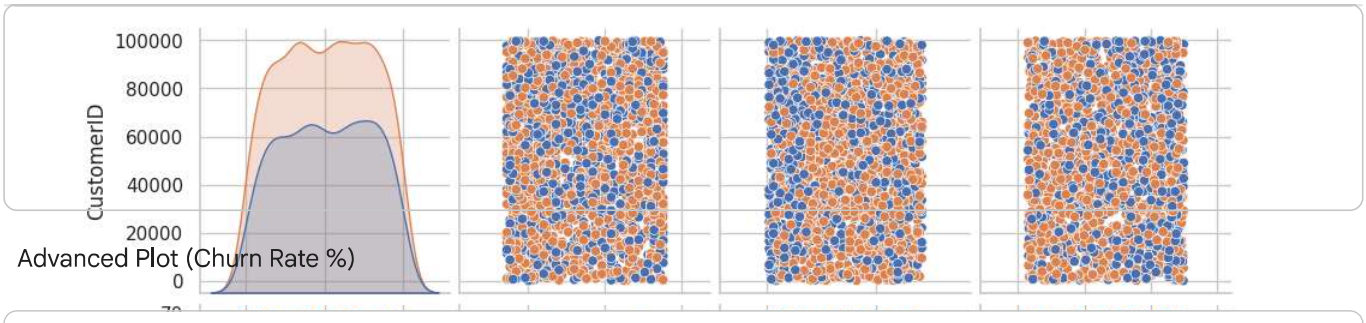

```
sns.countplot(x="City", hue="Churn", data=df)
plt.title("City-wise Customer Churn")
plt.xticks(rotation=30)
plt.show()
```

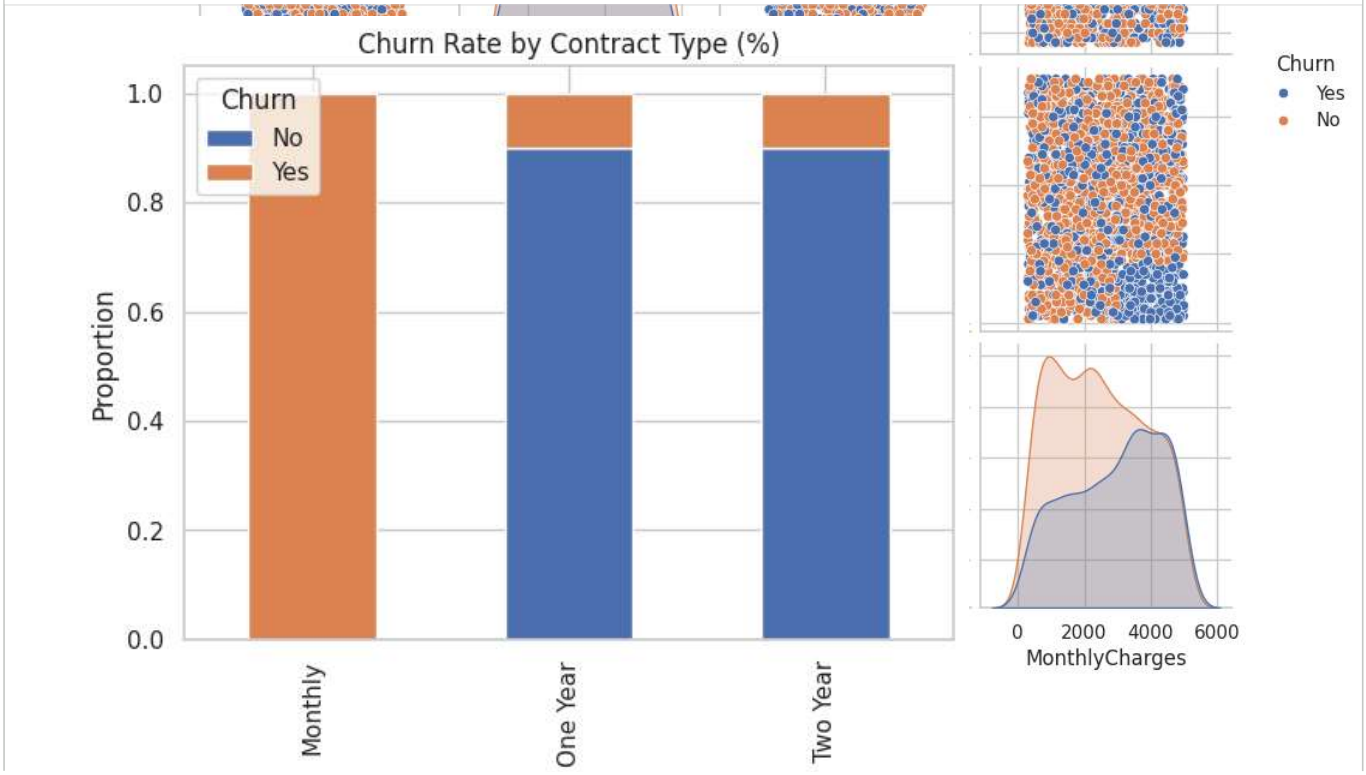



```
corr = df[["Age", "Tenure", "MonthlyCharges"]].corr()

sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
```




```
sample_df = df.sample(2000)

sns.pairplot(sample_df, hue="Churn")
plt.show()
```



Advanced Plot (Churn Rate %)

```
churn_rate = df.groupby("ContractType")["Churn"].value_counts(normalize=True).unstack()

churn_rate.plot(kind="bar", stacked=True)
plt.title("Churn Rate by Contract Type (%)")
plt.ylabel("Proportion")
plt.show()
```